

# On the Synergy of Temporal Sparsity and Local Observables: The Period-Locality Manifold Extension 7

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## Abstract

The optimal deployment of the Q-LINK architecture requires tuning both the finite-depth convergence dynamics and the asymptotic variance bounds. This note formally investigates Extension 7: the integration of a  $p$ -periodic residual messenger channel with an  $m$ -local averaged cost operator. By evaluating the Dynamical Lie Algebra (DLA) alongside Hilbert-Schmidt trace combinatorics, we prove that temporal sparsity strictly preserves the  $m$ -local trace advantage. This establishes a two-dimensional hyperparameter manifold  $(p, m)$  where the trajectory variance window and the global variance floor can be independently tuned, providing a highly elastic framework for barren plateau mitigation.

## 1 Orthogonality of DLA and Trace Normalization

Let the architecture be defined by a residual messenger connection applied every  $p$  layers, evaluated against an  $m$ -local averaged cost operator  $O_m$ .

As established in prior analysis of temporal sparsity, the continuous algebraic closure of the parameterized generators is invariant under periodic application. For any integer period  $p \geq 1$ :

$$\mathfrak{g}_p = \langle \mathcal{G}_{opr} \cup \mathcal{G}_{res} \rangle_{Lie} = \mathfrak{su}(2^{n+1}) \quad (1)$$

Because the  $p$ -periodic circuit asymptotically generates the exact same continuous sub-manifold as the dense circuit, the Haar integration volume evaluated by the Master Formula ( $U(2^{n+1})$ ) is strictly invariant with respect to  $p$ .

Simultaneously, the traceless part of the  $m$ -local cost operator  $\tilde{V}_m$  acts exclusively on the  $n$  data qubits. Its squared Hilbert-Schmidt trace norm depends only on the spatial locality  $m$  and the dimension of the extended Hilbert space, independent of the layer frequency:

$$Tr(\tilde{V}_{m,p}^2) = \frac{2N}{\binom{n}{m}} \quad (2)$$

## 2 The Asymptotic Bivariate Variance Bound

To evaluate the asymptotic advantage at Haar saturation, we compare the  $(p, m)$ -parameterized Q-LINK circuit against a Vanilla circuit operating on  $n + 1$  total qubits and evaluated with an equivalent  $(n + 1)$ -register  $m$ -local cost operator.

Because the DLA saturates the same Haar measure, the denominator of the Master Formula cancels. The generalized asymptotic variance ratio  $\mathcal{R}_{p,m}(n)$  is derived strictly from the ratio of the traces:

$$\mathcal{R}_{p,m}(n) = \frac{Tr(\tilde{V}_{m,p}^2)}{Tr(V_{Vanilla,m}^2)} = \frac{\frac{2N}{\binom{n}{m}}}{\frac{2N}{\binom{n+1}{m}}} = \frac{\binom{n+1}{m}}{\binom{n}{m}} \quad (3)$$

Expanding the binomial coefficients, we prove that the relative asymptotic advantage is strictly independent of  $p$ :

$$\mathcal{R}_{p,m}(n) = \frac{n+1}{n+1-m} \quad (4)$$

**Theorem of Synergy:** The combinatorial variance shield provided by highly non-local cost functions (large  $m$ ) is mathematically immune to the temporal sparsity of the residual channel (large  $p$ ).

### 3 The $(p, m)$ Optimization Manifold

While equation (4) dictates the deep-circuit geometric landscape, practical trainability is a finite-depth phenomenon. The parameters  $p$  and  $m$  act as independent, orthogonal controls for the optimization dynamics:

- **The Geometric Floor ( $m$ ):** The locality parameter  $m$  governs the absolute minimum magnitude of the gradient at Haar saturation. By shifting  $m \rightarrow n$ , the combinatorial trace advantage shields the gradient from the volumetric expansion of the messenger qubit.
- **The Trajectory Window ( $p$ ):** The period  $p$  governs the rate of operator spreading. A larger  $p$  fractionates the causal lightcone of the residual channel, throttling  $t$ -design convergence. The depth required to trigger the barren plateau scales proportionally to  $p \times D_{dense}$ , artificially widening the finite-depth optimization window.

### 4 Conclusion

Extension 7 establishes the most robust implementation of the Q-LINK architecture. By decoupling the temporal spread of correlations from the spatial normalization of the observables, the  $(p, m)$ -framework allows an optimizer to navigate an artificially extended trajectory variance window without sacrificing the mathematically guaranteed topological superiority of the network’s asymptotic geometry.